CASE STUDY 1: Examination Result

Python Activity-Based Learning

1. Problem Statement

Write a Python program to read three subject marks, calculate the total and average, and display whether the student passes or fails. A student passes only when every subject mark is at least 35.

2. Objectives

• Read marks for three subjects.

• Calculate total and average marks.

• Check the pass condition and display the result.

3. Assumptions

• Each subject mark is between 0 and 100.

• The pass mark in every subject is 35.

4. Algorithm

1. Start.

2. Read marks m1, m2 and m3.

3. Calculate total = m1 + m2 + m3.

4. Calculate average = total / 3.

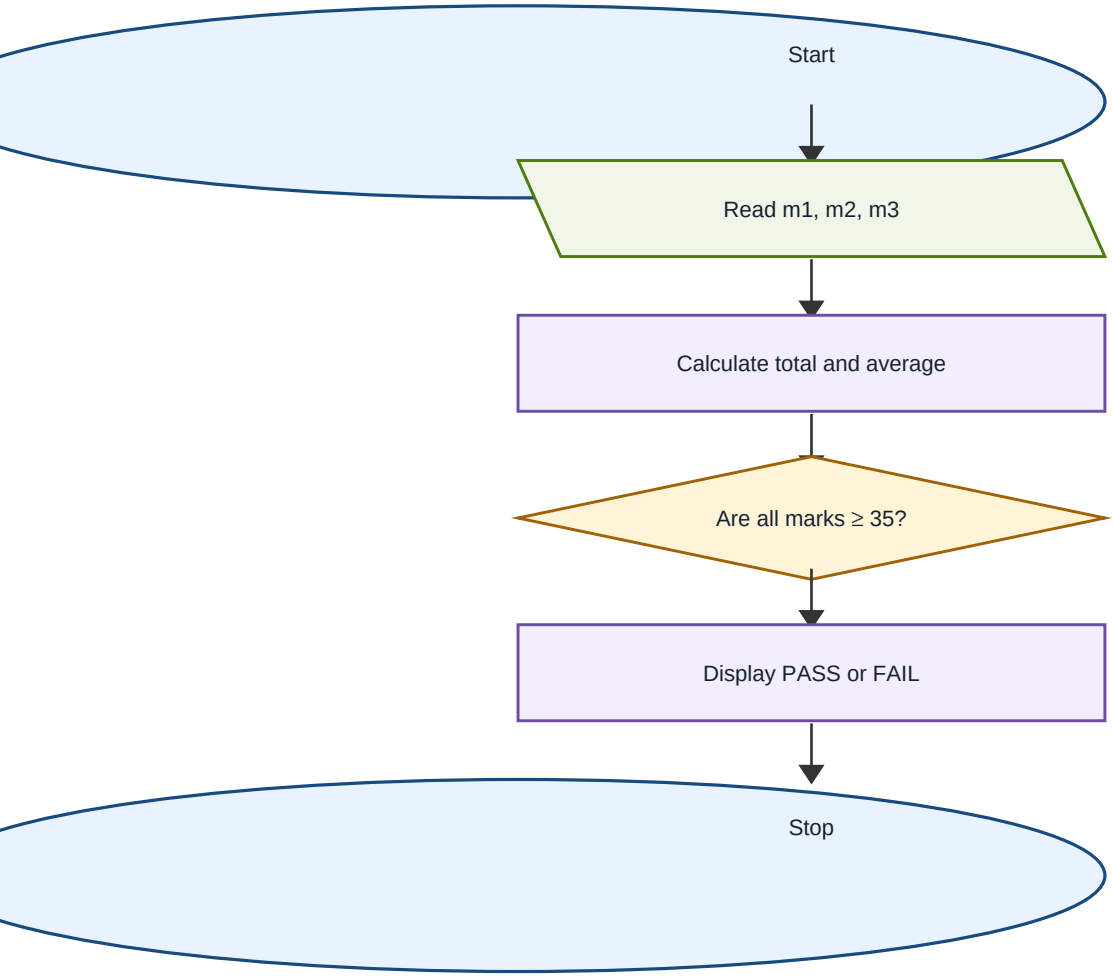
5. If all three marks are at least 35, display PASS; otherwise display FAIL.

6. Display total and average.

7. Stop.

5. Flowchart

Oval = Start/Stop; parallelogram = Input; rectangle = Process; diamond = Decision.



6. Pseudocode

```
BEGIN
  INPUT m1, m2, m3
  total = m1 + m2 + m3
  average = total / 3
  IF m1 >= 35 AND m2 >= 35 AND m3 >= 35 THEN
      PRINT "PASS"
  ELSE
      PRINT "FAIL"
  END IF
  PRINT total, average
END
```

CASE STUDY 1: Examination Result

7. Python Source Code

```
m1 = float(input("Enter marks in subject 1: "))
m2 = float(input("Enter marks in subject 2: "))
m3 = float(input("Enter marks in subject 3: "))

total = m1 + m2 + m3
average = total / 3

print("Total:", total)
print("Average:", average)

if m1 >= 35 and m2 >= 35 and m3 >= 35:
    print("Result: PASS")
else:
    print("Result: FAIL")
```

8. Test Cases

Input	Expected Output
60, 70, 80	Total = 210; Average = 70.0; PASS
30, 75, 80	Total = 185; Average = 61.67; FAIL

9. Sample Output

Run the Python program and enter the values from one of the test cases above. The displayed result should match the expected output.

10. Conclusion

The program calculates the total and average and uses a conditional statement to decide whether the student passes all three subjects.